

1. Suppose that a Web server runs in host C on port 80 using Http 1.1 protocol with persistent connections. Thus the Web server uses TCP transport service, and is currently receiving requests from two different hosts, A and B. Are all of the requests being sent through the same socket at host C? If they are being passed through different sockets, do both of the sockets have port 80? Discuss and explain.

ANS: 並非使用相同的 Socket, 因為不同的來源 IP 會被視為不同 Socket
但兩個連線皆是傳入 PORT 80, 因為 Host C 在 PORT 80 提供服務

2. Suppose Host A sends two TCP segments back to back to Host B over a TCP connection. The first segment has sequence number 90; the second has sequence number 110.

- a. Suppose that the first segment is lost but the second segment arrives at B. In the acknowledgment that Host B sends to Host A, what will be the acknowledgment number?

ANS: 90

- b. How much data is in the first segment?

$$110 - 90 = 20$$

ANS: 20 Bytes

3. Consider the GBN protocol with a sender window size of 3 (packets) and a sequence number range of 1,024. Suppose that at time t , the next in-order packet that the receiver is expecting has a sequence number of k . Assume that the medium does not reorder messages. Then, what are the possible sets of sequence numbers inside the sender's window at time t ? Justify your answer.

ANS: $(k, k+1, k+2) \dots$ Sender 收到 ACK ($k-1$)

$(k-1, k, k+1) \dots$ Sender 收到 ACK ($k-2$) 但未收到 ACK ($k-1$)

$(k-2, k-1, k) \dots$ Sender 收到 ACK ($k-3$) 但未收到 ACK ($k-2$)

$(k-3, k-2, k-1) \dots$ Sender 收到 ACK ($k-4$) 但未收到 ACK ($k-3$)

→ 沒有其他情況, 因為接收端期望 k 時表示 Sender 至少送出了 $k-1$,

所以不會出現 $(k-4, k-3, k-2)$ 的情況

4. Host A and B are directly connected with a 200 Mbps link. There is one TCP connection between the two hosts, and Host A is sending to Host B an enormous file over this connection. Host A can send application data into the link at 100 Mbps but Host B can read out of its TCP receive buffer at a maximum rate of 50 Mbps. Describe the effect of TCP flow control.

ANS: Host B 會在給 Host A 的 ACK 封包中附上自己的 receive window 的剩餘空間，一開始 Host B 的 Buffer 會被快連佔滿，而 Host A 檢測到後會降低發送速度，最終 Host A 的發送速度也會被限制在 50 Mbps

5. For TCP's estimation of RTT. Why do you think TCP avoids measuring the **SampleRTT** for retransmitted segments?

ANS: 因為我們無法從封包中得知這是第幾次重傳，也就無法得知一次封包來回的正確時間